

AI and GenAI Assignment - Report

1. Objective

The objective of this assignment is to implement a simple Generative Adversarial Network (GAN) using TensorFlow and the MNIST dataset. The goal is to generate synthetic handwritten digit images that resemble real MNIST digits.

2. Brief Explanation of GAN

A Generative Adversarial Network consists of two neural networks:

Generator

- Takes random noise as input.
- Produces synthetic images.
- **Goal:** fool the discriminator.

Discriminator

- Receives real and fake images.
- Classifies them as real (1) or fake (0).
- **Goal:** correctly detect fake images.

Both networks compete in a **minimax game**:

- Generator improves to fool discriminator.
 - Discriminator improves to detect fakes.
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3. Architecture Summary

Generator

Layers:

1. Dense layer
2. Batch Normalization
3. LeakyReLU
4. Conv2DTranspose layers
5. Output with Tanh activation

Output: 28×28 grayscale image.

Discriminator

Layers:

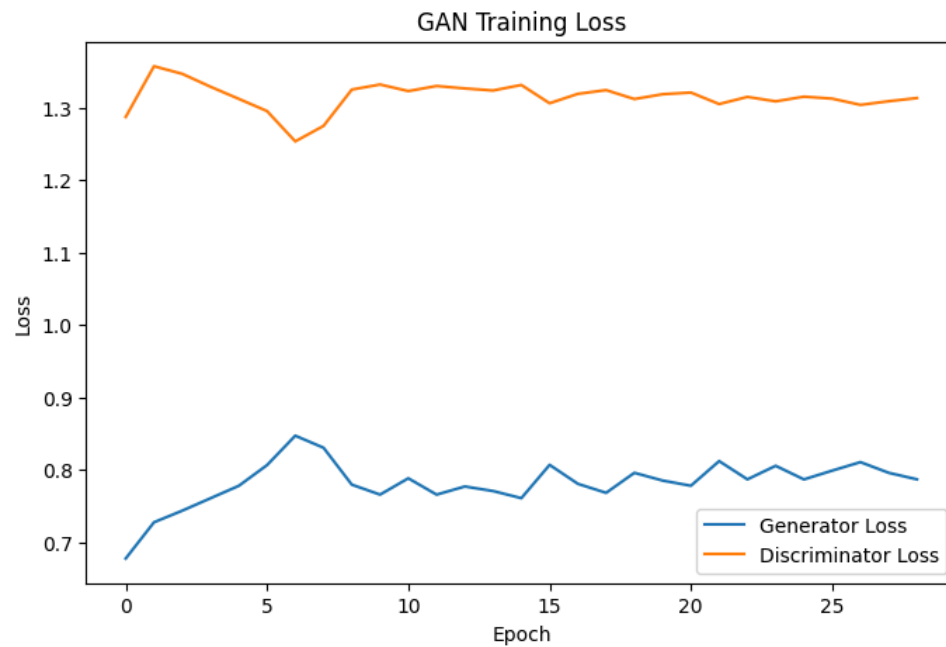
1. Conv2D
2. LeakyReLU
3. Dropout
4. Flatten
5. Dense layer with sigmoid

Output: Probability (real or fake).

4. Training Details

- Dataset: MNIST
 - Normalization: [-1, 1]
 - Loss: Binary Cross-Entropy
 - Optimizer: Adam (lr=0.0002, $\beta_1=0.5$)
 - Epochs: 20
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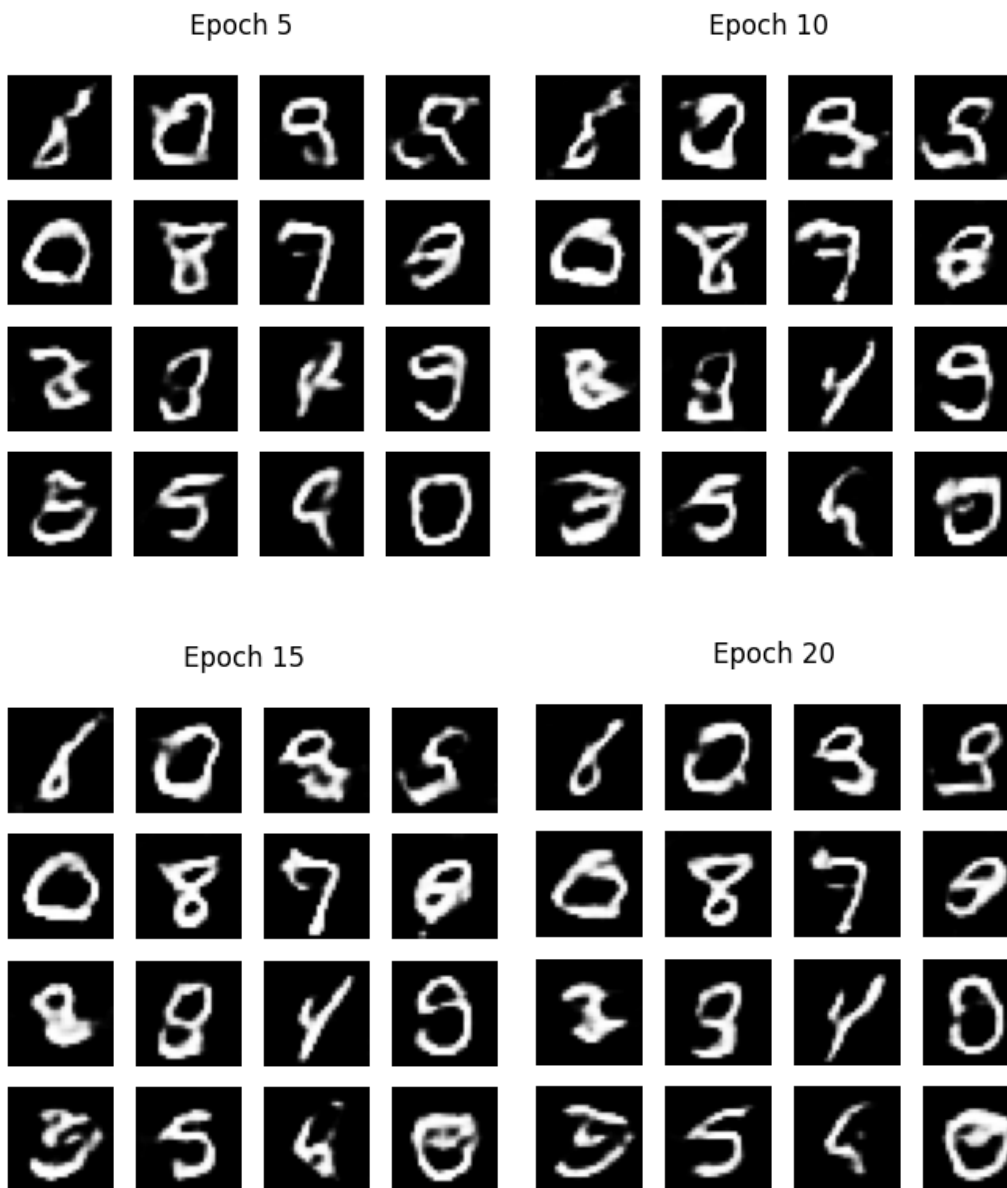
5. Loss Graph



Observation:

The generator and discriminator losses fluctuate throughout training rather than decreasing monotonically. This is expected behavior in GANs because both networks are trained in an adversarial setting. The generator loss stabilizes around 0.75–0.82, while the discriminator loss remains around 1.28–1.33, indicating balanced training. The generator gradually improves in producing realistic digit images.

6. Sample Generated Images



Observation:

- Early epochs: noisy shapes.
 - Later epochs: clearer digit structures.
 - The generator gradually improves realism.
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7. Challenges Faced

- Training instability in GANs.
 - Loss values fluctuate instead of decreasing.
 - Balancing generator and discriminator learning rates.
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8. Conclusion

The GAN successfully generated handwritten digit images similar to the MNIST dataset. The adversarial training helped the generator improve over epochs, producing increasingly realistic digits.